Rational Choice Thoery

Spring 2026 (Block IV)

Dr. Babak RezaeeDaryakenari

Leiden University
The Institute of Political Science

Aggregating preferences and problems it raises

- Quick review of previous sessions
- Social Choice Theory
- Cyclical majority and Condorcet paradox
- Arrow Impossibility Theorem
- May's theorem
- Single-Peakedness
- Median Voter Theorem

Quick review of last three weeks

- Rational choice, utility, and expected utility for individuals
- The theories of games and strategic interactions
- The failure of *classic* rational choice models in explaining the provision collective goods
- A behavioral model of choices

Social Choice Theory: another case for violating rational choice assumptions

- Last week, we discussed some challenges that rational choice theory faces!
- This week: we discuss another common social phenomenon that challenges rational choice assumptions:
  - The aggregation of individual preferences into social preferences
  - Studying this aggregation and its implications for individuals and society is called ***social choice theory***.
- The main question is whether aggregating rational individuals' preference lead to the violation of rational choice properties?
- If yes, then what the **proposed solutions** are!

As always: Let's start with an example

Consider a simplified group choice example:

- Assume the university reaches a group of 3 representative BA students to ask them about next year's teaching plans
- $Students = \{Abby(A), Bob(B), Camilla(C)\}$
- $Methods = \{In\ person(IP), Online(ON), Hybrid(HY)\}$
- Each of these students have their preferences ordered over these options
- How can the students aggregate their preferences? That is, reach a group decision/choice.

Students' preferences: Version 1

- Abby: $IP \succ_A ON \succ_A HY$
- Bob: $IP \succ_B HY \succ_B ON$
- Camilla: $IP \succ_C ON \succ_C HY$

Under this situation, finding the group's preferred option is easy: IP
However, this is not the case always.

Students' preferences: Version 2

- Abby: $IP \succ_A ON \succ_A HY$
- Bob: $ON \succ_B HY \succ_B IP$
- Camilla: $HY \succ_C ON \succ_C IP$

Abby	Bob	Camilla
IP	ON	HY
ON	HY	ON
HY	IP	IP

- Under these preferences, finding the group's preferred choice is not straightforward!
- We need a rule to reach an agreement about group choice. These types of rules are called *majority rule*.

Majority rule: Rule for aggregating individuals

"These smug pilots have lost touch with regular passengers like us. Who thinks I should fly the plane?"

Source: The New Yorker

Students' preferences: Version 2

Abby	Bob	Camilla
IP	ON	HY
ON	HY	ON
HY	IP	IP

- Rule: Round robin tournament
  - Members vote on all options pairwise
  - Group choice: The choice preferred by a majority to all the others

IP vs. ON	ON wins (2-1)	ON $\succ_G$ IP
IP vs. HY	HY wins(2-1)	HY $\succ_G$IP
ON vs. HY	ON wins (2-1)	ON $\succ_G$ HY

Group choice/ranking: $ON \succ_G HY \succ_G IP$

Students' preferences: Version 3

- The preference orders of members can also lead to situations that round robin tournament fails as majority rule.
- Let's look at another example.
- Abby: $IP \succ_A ON \succ_A HY$
- Bob: $ON \succ_B HY \succ_B IP$
- Camilla: $HY \succ_C IP \succ_C ON$

Abby	Bob	Camilla
IP	ON	HY
ON	HY	IP
HY	IP	ON

Students' preferences: Version 3

Abby	Bob	Camilla
IP	ON	HY
ON	HY	IP
HY	IP	ON

- Rule: round robin tournament:
  - Members vote on all options pairwise
  - Group choice: The choice preferred by a majority to all the others

IP vs. ON	IP wins (2-1)	IP $\succ_G$ ON
IP vs. HY	HY wins(2-1)	HY $\succ_G$IP
ON vs. HY	On wins (2-1)	ON $\succ_G$ HY

- Group choice/ranking: $IP \succ_G ON \succ_G HY \succ_G IP$
- What is the problem with this ranking?Poll!

Poll

Which one of the below is correct about
$IP \succ_G ON \succ_G HY \succ_G IP$?

- The order violates the completeness property of rational choice.
- The order violates the transitivity property of rational choice.
- The order violates both completeness and transitivity.
- The order does not violate rational choice properties.

Cyclic majority

Definition: *Cyclic majority*

An *intransitive preference* order arising from majority voting in a group of individuals with *transitive individual* preferences.

- This paradox is called *Condorcet Paradox*, *voting paradox*, or *the paradox of voting*.
- It was modeled by Lewis Carroll and Edward J. Nanson,
- but Duncan Black made it popular in Social Choice Theory.

Arrow's impossibility theorem

- The most well-known Social Choice Theory.
- Considering different preferences that we discussed A fundamental question: Is there a rational and fair majority rule that is democratic?
- Arrow's answer is no!

Theory: *Arrow's Impossibility Theorem*

Suppose that there are at least three candidates (alternatives/choices) and finitely many voters. Any *social welfare function* that satisfies *universal domain*, *irrelevance of independent alternatives*, and *unanimity* is a *dictatorship*.[1]

[1] Arrow, Kenneth J. Social choice and individual values. John Wiley & Sons, 1951.

Arrow's Impossibility Theorem (cont'd)

- The key concepts in Arrow's theorem:
  - Social welfare function
  - Universal domain
  - Irrelevance of independent alternatives
  - Unanimity
  - Dictatorship
- Universal domain, Irrelevance of independent alternatives, and unanimity represent the conditions for a consist *reflection* of *all* individual preferences in social welfare/aggregation function

Arrow's Impossibility Theorem (cont'd)

Definition: *Social Welfare Function*

A social welfare function, or social welfare aggregator, is a rule $F(\nu_1, \nu_2, \ldots, \nu_I)$ that assigns a social preference, $F(\nu_1, \nu_2, \ldots, \nu_I)$={-1,0,1}, to every possible profile of individual preferences $(\nu_1, \nu_2, \ldots, \nu_I) = \{-1, 0, 1\}^I$

- For individual i who is offered to decide between x and y:
  - $\nu_i = 1$ *if* $x \succ_i y$
  - $\nu_i = 0$ *if* $x \sim_i y$
  - $\nu_i = -1$ *if* $y \succ_i x$
- For a society of I individuals:
  - What does $(\nu_1 = 1, \nu_2 = 1, \nu_3 = 0, \nu_4 = -1, \ldots, \nu_I = -1)$ mean?
  - For the sake of simplicity: $(\nu_1 = 1, \nu_2 = 1, \nu_3 = 0, \nu_4 = -1, \ldots, \nu_I = 1) = (1, 1, 0, -1, \ldots, -1)$

Arrow's Impossibility Theorem (cont'd)

Definition: *Unanimity (Condition U), also known as Pareto Optimality*

If each individual ranks x above(below/indifferent) y, then the social function also does this ranking.

- Examples:
  - All individuals rank x above y $(1, 1, 1, \ldots, 1)$, then $F(1, 1, \ldots, 1) = 1$
  - All individuals indifferent between x and y $(0, 0, 0, \ldots, 0)$, then $F(0, 0, \ldots, 0) = 0$
  - All individuals rank y above x $(-1, -1, -1, \ldots, -1)$, then $F(-1, -1, \ldots, -1) = -1$

Arrow's Impossibility Theorem (cont'd)

Definition: *Universal domain*

Individuals are free to choose any preference they want.

- This means if there are two alternatives x and y: Then the available rankings are: $x \succ_i y$, $y \succ_i x$, or $x \sim_i y$
- For three alternatives x, y, and z, this means:

1	2	3	4	5	6	7	8	9	10	11	12	13
x	x	y	y	z	z	xy	xz	yz	x	y	z	
y	z	x	z	x	y							x y z
z	y	z	x	y	x	z	y	x	y z	x z	xy	

Arrow's Impossibility Theorem (cont'd)

Definition: *Independence of Irrelevant Alternatives*

The social ranking of two alternatives x and y depends only on how each individual in the society rank x and y.

- This means if the social welfare function ranks, for instance, x above y ($x \succ_S y$), then adding alternative z to the alternatives/choice set or considering it, does not change the social order of x nd y.
- Example: $\{Coffee, Tea\}$: $Coffee \succ_S Tea$
- $\{Coffee, Tea, Juice\}$: Still $Coffee \succ_S Tea$, not $Tea \succ_S Coffee$ nor $Tea \sim_S Coffee$

Arrow's Impossibility Theorem (cont'd)

Definition: *Dictatorship*

An individual D from society is a dictator if society ranks x over(below, indifferent) B whenever D ranks x over(below, indifferent) y.

- That is, in extreme cases, even if all individuals vote similar to each other, but different from the dictator, then dictator's preference is picked.
  - $F(1, 1, \ldots, 1, \nu_D = -1, 1, \ldots, 1) = -1$
  - Or, $F(-1, -1, \ldots, -1, \nu_D = 1, -1, \ldots, -1) = 1$
  - Or, $F(-1, -1, \ldots, -1, \nu_D = 0, -1, \ldots, -1) = 0$
  - And, so forth

Arrow's impossibility theorem: Summary

Theory: *Arrow's impossibility*

Suppose that there are at least three candidates (alternatives/choices) and finitely many voters. Any *social welfare function* that satisfies *universal domain*, *irrelevance of independent alternatives*, and *unanimity* is a *dictatorship*.

- Arrow mathematically proves: for a finite number of voters with more than three alternatives there is always a dictator whose vote is prioritized to other voters!
- IMPORTANT: No need to learn the proof of Arrow's Impossibility Theorem.[2]
- Just digest the implication: when aggregating rational individuals' preference, it is not democratic, and there is one dictator!

[2] But, if interested to take a look. Here it is: See

May's Theorem

- You might ask, what if there are only two candidates.
- If there are two candidates:
- May's theorem: a majority rule that picks the candidate with largest share of votes (counting noses) by rational voters if and only if four May's conditions are satisfied.
- May's conditions are special cases of Arrow's condition.
- Therefore, Arrow's argument, incoherent group preferences, also applies to two candidate's case with the method of majority rule of counting noses.

Some clarifications about Arrow's Impossibility Theorem

- Arrow's finding can be disturbing if we oversimplify his theorem and conclude: "Democracy is impossible!"
- Arrow's main argument is that we should not expect a collective of individuals *always* to behave like a rational individual!
- Arrow's theorem, in essence, underlines the importance of institutions and the rules that we impose to aggregate individuals' decisions into a collective decision!

Single-Peaked preferences

- Arrow's and May's theorems challenge:
  - the value of majority rule for aggregating rational individuals' preferences into a coherent preferences at the group/society level.
- However, rational individuals/agents preferences are not always discrete with challenging properties.
- Individuals can have preferences with monotone[3] and nice properties.

[3]Monotone preference roughly means there is smooth increase or decrease in ranking of alternatives.

An famous example of well-behaved preferences

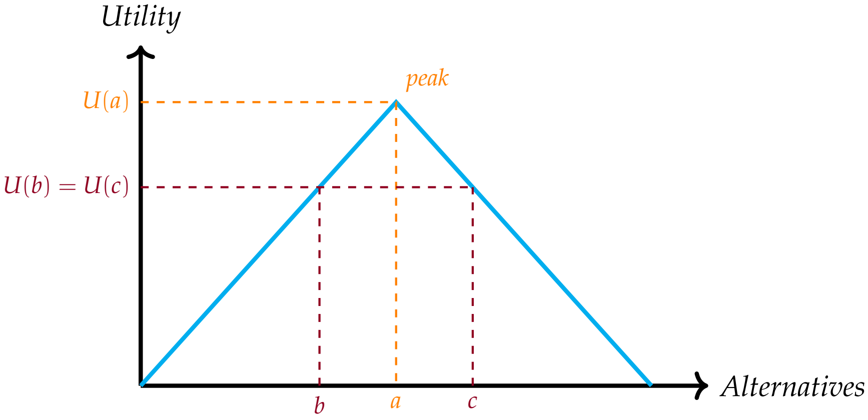
Definition: *Single-Peaked Preferences*

Preferences are said to be single-peaked if the alternatives can be represented as points on a line, and each utility function has a maximum at some point on the line and slopes away from the maximum on either side.

Single-Peaked preferences: Example

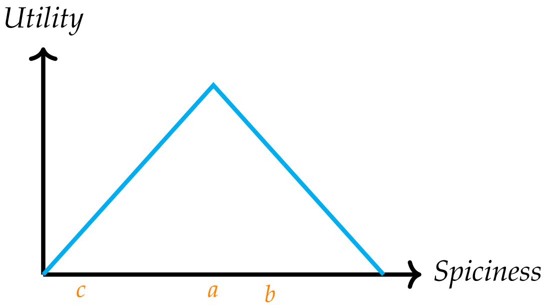
- When the preferences are ranked from left to right over a continuous line.
  - Your preference over your grade.
  - Preference over candidates in an election between left and right spectrum
  - The preferred level of repression by security forces
  - The preferred level of tax
  - The preferred level of violence by protesters
- In all of these cases, individuals have a most-preferred alternative/option/candidate, and then moving toward extremes decreases their utility toward less preferred alternatives.

Single peaked preferences



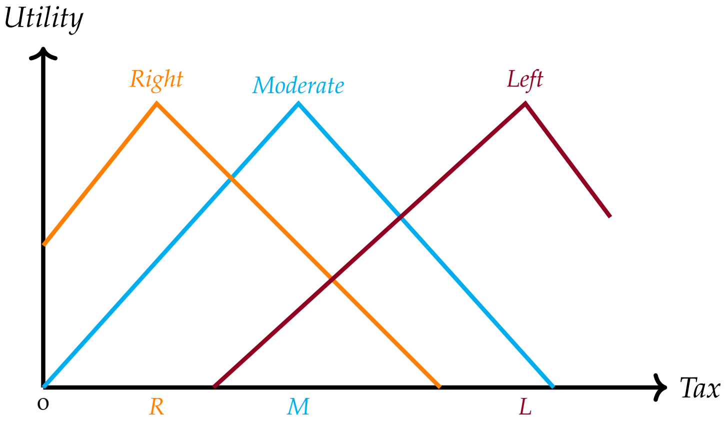
- Therefore, $U(a) > U(b) = U(c)$, or
- $a \succ b \sim c$

Single peaked preferences: Example



- Assume the above utility function shows Bob's preference over spicy food. How do you rank his preferences? Poll!
    A. $a \succ b \succ c$
    B. $a \succ c \sim b$
    C. $b \succ a \sim c$
    D. $c \succ a \sim b$

Single peaked preferences: Left, right, moderate



Median voter theorem

- Median peak is the winner of majority rule of round-robin tournament, also known as Condorcet method.
- This single-peaked preferences is used by Duncan Black to develop Median Voter Theorem:

Median Voter Theorem

If preferences are single-peaked and uniformly distributed over the alternatives line, the closest alternative to median alternative is chosen by the group/society.

Take away from this course

- Rational choice and/?or formal mathematical
- Rational choice models: simplified but powerful tool to study social phenomenon.
- One advantage of formal rational choice: being specific and explicit about its assumptions
- Rational choice models: private goods/outcomes vs. collective goods/outcomes
- Behavioral models of rational choice: Human is not perfect!
- Social choice theory: Be careful when aggregating individual preferences

Last words

- Our world is too complex: There is not one-size-fits-all solution
- Be literally and genuinely open to all methods and thoughts!
- While you learned about positivist rational choice, with an emphasize on mathematical modeling, be open to other methods and approaches such as Critical Theory!
- Most of famous mathematicians were famous philosophers, too!
- Many of school of thoughts have more overlap than differences! Example:
  - The foundation of neoclassic growth model is based on Marx's conceptualization of capital accumulation: $k_{t+1} = (1 - \delta)k_t + S_t$!
- **Rational choice** is **neither** the **best nor** the **worse** method to answer social science questions. *It is a tool like other tools in your toolbox! Use it wisely!*